Key Considerations for re-qualification of Pipelines for Carbon Dioxide in Gas Phase transport

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1 ABSTRACT

In the context of carbon capture utilization and storage (CCUS), reuse of existing onshore and offshore pipelines may be an attractive alternative to installing new pipelines. Through different work packages, the ongoing CO2SafePipe Joint Industry Project (JIP) addresses identified challenges when repurposing pipelines for the transport of carbon dioxide in both gas and dense phases. This initiative aims to ensure that a repurposed pipeline adheres to the same stringent safety and performance standards as new pipelines. Dense phase condition has historically been the preferred option for pipelines specifically designed for transporting larger quantities of carbon dioxide. When considering the reuse of existing pipelines for transporting carbon dioxide, dense phase may not always be a technically viable option due to limitations from the original design of the pipeline planned for reuse. This paper delves into the specific challenges of re-qualifying pipelines for transporting carbon dioxide in gas phase, identifying key considerations, opportunities and limitations. Special attention is given to pipeline operating envelope, product composition, material properties, integrity management, and safety considerations.

2 INTRODUCTION

Carbon Capture and Storage (CCS) value chains rely on the transport of carbon dioxide (CO_2) from the location of capture to the location of storage in a safe and cost-effective manner. Pipelines have to date been the preferred option for the transport of larger volumes across moderate to long distances. Ship transport of CO_2 is currently limited to smaller ship sizes with small to moderate transport capacities. It is foreseen that CO_2 carriers with larger transport capacities will be developed in the future to meet the demands of a large-scale CCS market. Hence, the transport of CO_2 in the context of CCS is expected to include both ship and pipeline transport within different parts of the value chain. The preferred options will be a function of main drivers such as transport volumes, transport distance, need for continuous transport and the expected design life of the transport value chain. For onshore transport of larger volumes, pipelines will likely be the preferred option compared to truck, rail or barge transport. Repurposing both onshore and submarine pipelines may be an attractive option to reduce CO_2 pipeline investment costs. The potential cost saving of reuse of pipelines versus newbuild depends on the required pipeline modification work to conclude compliance.

Over the recent years, several pipeline operators have been considering repurposing existing pipelines for CO_2 transport. For the design of new pipelines, it is an established opinion that transportation of CO_2 in dense phase rather than gas phase is the preferred option. However, several re-qualification studies point to the transportation of CO_2 in gas phase as an attractive option. In a comprehensive study initiated by IOGP[1] (2021) /3/, it was concluded that a large fraction of existing oil and gas pipelines within Europe would be technically suitable for transporting carbon dioxide in gas phase, and only a small fraction suitable for dense phase. The reason for this limitation is mainly due to insufficient design pressure or insufficient capacity to arrest a running fracture for dense phase operation. The study covers both onshore and offshore pipelines. It should be acknowledged that this study was based on a high-level screening and that individual pipelines will need to be subject to a pipeline specific re-qualification process for final confirmation of suitability, under what premises and with what required modifications.

This paper addresses current experience with repurposing pipelines for the transport of carbon dioxide in gas phase and the key considerations, limitations and opportunities relative to dense phase. Typical steps to be followed for pipeline re-qualification are described in DNV-SE-0657 /4/, see Figure 2-1.

[1] International Association of Oil & Gas Producers

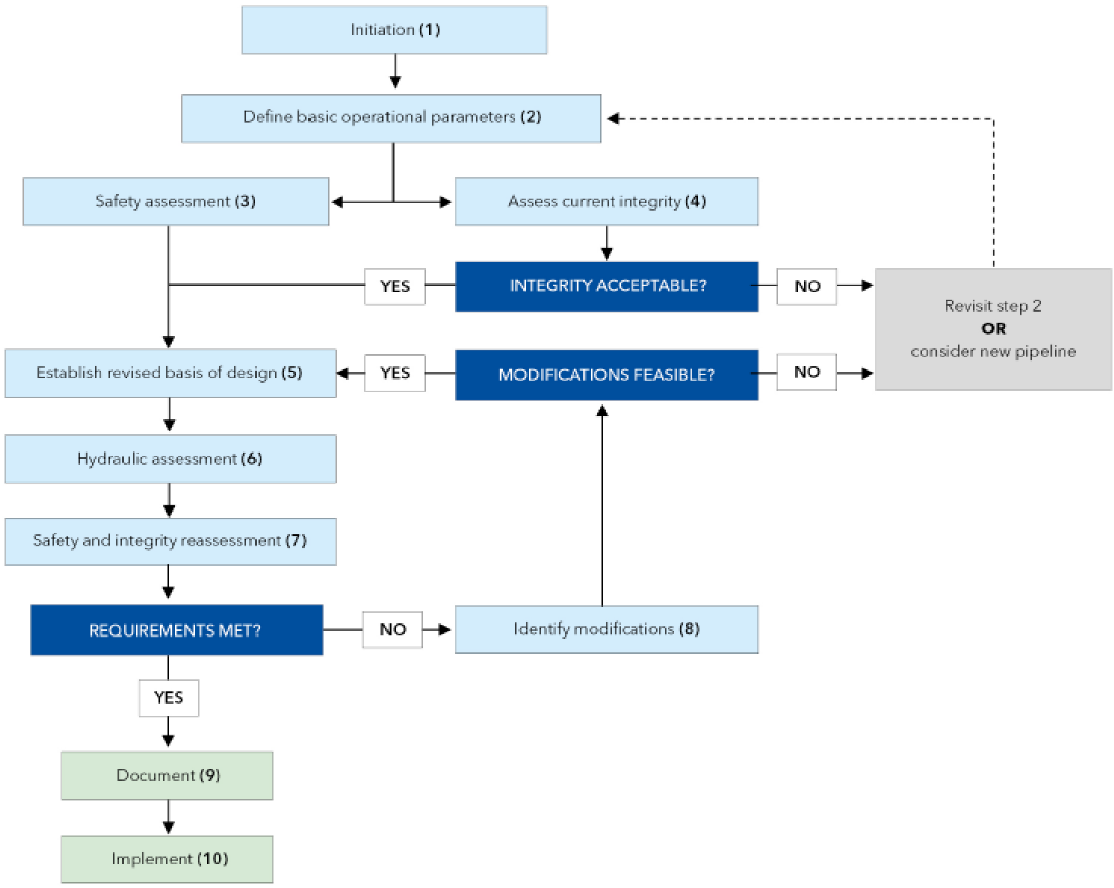

Figure 2-1 Ten-step process for the re-qualification of pipeline systems /3/ /15/

3 INDUSTRY EXPERIENCE

Industry experience with gas phase carbon dioxide pipelines is generally less than for dense phase pipelines and limited to moderate distance transport of moderate amounts of carbon dioxide for industrial use. Two examples of pipelines repurposed for the transport of carbon dioxide in gas phase are described in the following.

The OCAP (Organic Carbon dioxide for Assimilation in Plants, The Netherlands) project in Rotterdam utilizes an onshore vintage oil pipeline re-qualified for carbon dioxide. The carbon dioxide captured from different local emitters is delivered through an approximately 100 km long 26” pipeline to greenhouses to enhance crop growth /7/. The carbon dioxide is captured from the Shell Pernis refinery and, since 2010, from the Abengoa bioethanol plant. The pipeline system operates in gas phase at pressures in the range of 7 to 21 bara and ambient temperature. The pipeline has been successfully operated since 2005, providing approximately 0.5 million tonnes per annum (MTPA) of carbon dioxide to a large number of greenhouses within the larger Rotterdam area. Strict product composition control is imposed on water to manage internal corrosion and other impurities limited by the end user (specification for greenhouse).

The LACQ CCS project in South-West France was operated by TOTAL as a demonstrated pilot from 2010, capturing carbon dioxide from an oxy-fuel combustion process and transporting the carbon dioxide in an approximately 30 km (12”/8” diameter) long pipeline to the injection site /8/. The onshore pipeline was originally designed and operated for the transport of natural gas (40 years) and converted to the transport of carbon dioxide for the pilot project. For carbon dioxide, the pipeline flow direction was reversed relative to the original operation with natural gas. The pipeline successfully transported dehydrated carbon dioxide at an operating pressure of 27 bar and 30°C in gas phase for a test period of approximately three years, from January 2010 to March 2013.

The above examples demonstrate that reusing pipelines for transporting carbon dioxide in gas phase is feasible. In several ongoing CCS projects planning for injection to depleted oil & gas reservoirs, the low (depleted) initial reservoir pressure may also dictate the transport of carbon dioxide in gas phase until reservoir pressure has increased sufficiently to allow for dense phase conditions arriving at the injection site.

4 COMPARISON OF DENSE PHASE VS GAS PHASE

A high-level identification of possible pros and cons for repurposing pipelines to gas phase versus dense phase is summarized in Table 4-1. The list is non-exhaustive and should only be taken for guidance, differences may be governed by pipeline specific factors or local regulations.

On the negative side, the main challenge for gas phase operation is the limitation in transport capacity due to the density of gas phase being significantly less than dense/liquid phase and the limitation in maximum operating pressure to maintain a single gas phase. On the positive side, operation in gas phase is generally perceived as more comparable to natural gas pipelines with regards to safety risk which may ease permitting and cause fewer challenges with public acceptance. Required operating pressure is also within range of typical design pressure for onshore pipelines. For dense phase operation, high attention is given to documenting sufficient capacity to arrest a running fracture. For gas phase pipelines this challenge is significantly less and more comparable to natural gas pipelines, see section 8 for further guidance.

Table 4-1 Pros and cons of dense and gas phase operation; key considerations outlined in bold

	Dense phase	Gas phase
Pro	• **Transport capacity** • Applicable to pipelines with a lower diameter to wall thickness ratio (D/t) due to higher design pressure, increasing the general robustness • Higher tolerance to water content (higher solubility) • Operational experience (mainly dense/liquid)	• **Safety (consequence zone/radius)** • **Required design pressure** • Operability (comparable to gas pipelines) • Moderate effect of hydrostatic pressure (hydraulics) • Less sensitive to (high) pipeline inlet temperature • **Running ductile fracture** • Spacing of line break valves • Permitting (comparable to gas pipelines) • Public acceptance (comparable to gas pipelines)
Con	• **Safety (consequence zone/radius)** • **Required design pressure** • **Running ductile fracture (more challenging)** • Possible need for installing fracture arrestors • Low temperature effects (brittle fracture) • Operability constraints • Pipeline (segment) depressurization (duration, dispersion, noise, dry ice) • Required spacing of line break valves • Effect of topography on pressure profile (hydraulics) • Sensitivity to pipeline inlet temperature (transport capacity) • **Regulations (novel for Europe-populated areas)** • Public acceptance (novel for Europe-populated areas)	• **Transport capacity** • Limit on maximum operational pressure (affecting transport capacity • Lower water solubility (stricter water specification needed)

5 PIPELINE OPERATING ENVELOPE FOR GAS PHASE

Transporting carbon dioxide in gas phase is limited by the product composition phase envelope. Figure 5-1 shows a typical range in operating conditions, visualized for simplicity on a composition with 100% carbon dioxide. There is in principle no lower limit in operating pressure for gas phase carbon dioxide pipelines other than what is determined by the required pipeline transport capacity or minimum arrival pressure at the pipeline downstream end.

To avoid condensation and the formation of a two-phase flow, the maximum allowable operating pressure as a function of operating temperature needs to consider a sufficient margin to the condensation line. The required margin should be evaluated project specific, allowing for optimization of pipeline operation both for normal, transient and upset conditions. The margin should also consider the potential variation in product composition and the expected accuracy of any prediction models used to support pipeline operation. Operation close to the critical point (31°C / 74 bara for pure CO_2) is generally associated with higher model uncertainties and high gradients on key thermos-physical properties. Adding impurities to the composition generally results in the condensation line being lifted, i.e. a higher condensation pressure for a given operating temperature, see Figure 5-2. The inclusion of heavier (liquid) impurities may have the opposite effect of reducing the condensation pressure.

When deciding the maximum allowable operating pressure, the temperature at the pipeline inlet and distributed temperature along the pipeline length as a function of pipeline operation, thermal insulation and ambient conditions should be considered. The most conservative approach would be to set the maximum operating pressure with reference to the minimum ambient temperature. For shut-in and settle-out conditions, it shall normally be assumed that the product temperature may stabilize at the ambient soil/sea temperature. At typical 1.2 m (4 ft) pipeline burial depth onshore, lowest soil temperatures ("winter-Europe") are typically higher than 5-10°C. While, for offshore pipelines, minimum seabed temperature is normally higher than 0-5°C. I.e., for a typical reference ambient (winter) soil temperature of 10°C, maximum allowable pipeline operating pressure would need to be limited to typically 40 bara to leave a 5 bar margin to the condensation line. For reference, the Porthos 42" onshore gas phase pipeline is planned to be operated at 35 bar /*9*/.

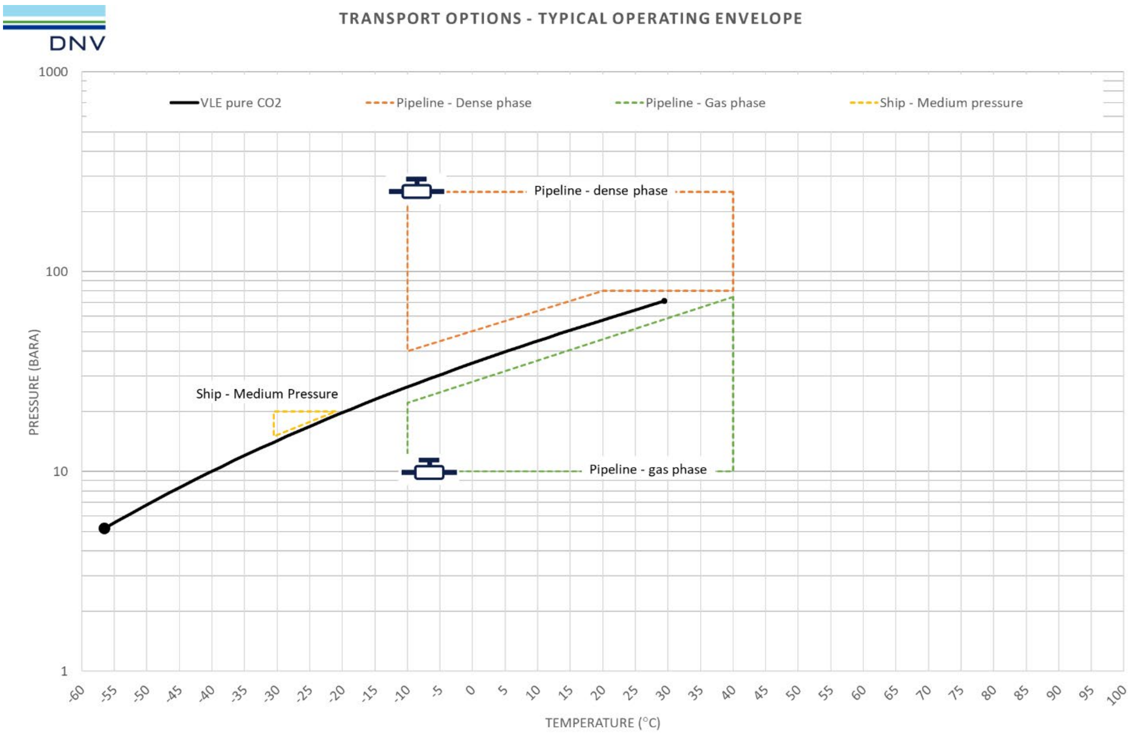

Figure 5-1 Phase envelope for 100% CO_2; typical transport conditions for gas and dense phase

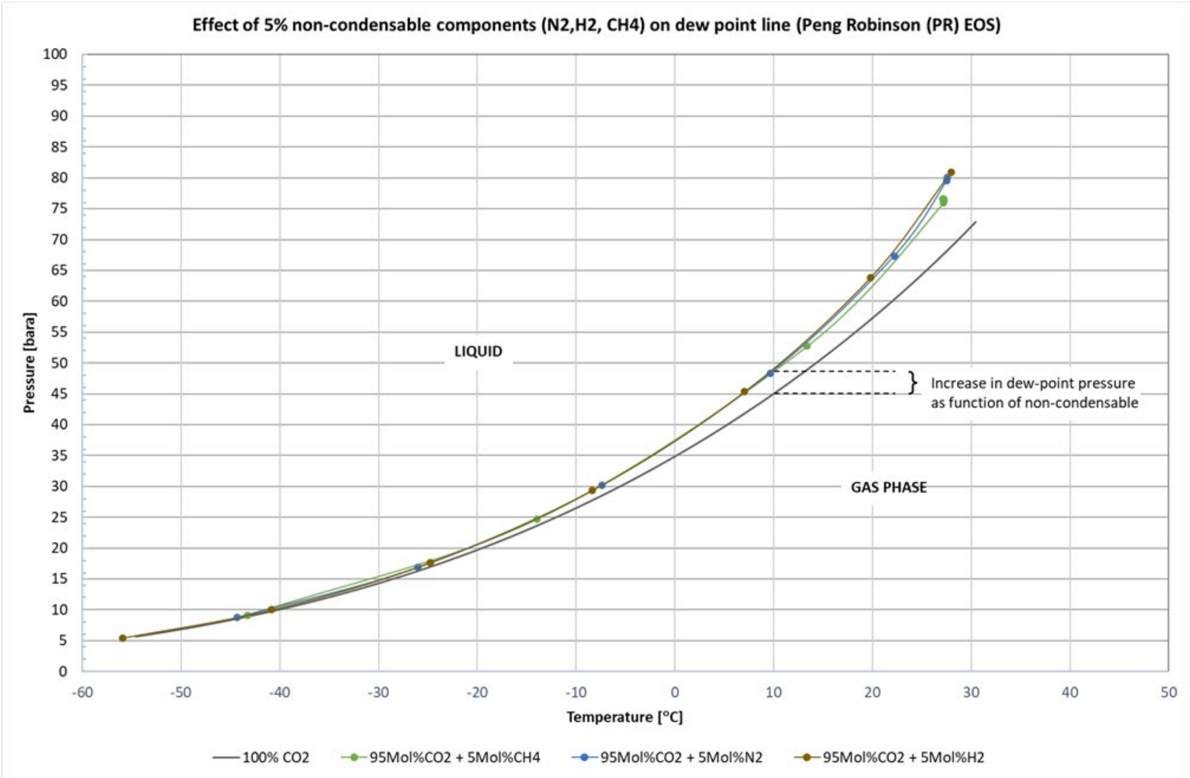

Figure 5-2 Dewpoint line for 100% CO2 and 95% CO2 with selected non-condensable components (5%)

For gas phase operation, the viscosity of carbon dioxide is comparable to natural gas, hence with a similar challenge related to pipeline friction loss. Also, considering the narrow envelope on operating pressure for gas phase carbon dioxide pipelines means reducing friction loss may be important. Re-qualifying pipelines with internal flow coating may therefore be perceived as attractive. Attention shall then be given to the qualification of the coating with regards to compatibility with CO_2 product composition, including the effects of impurities. For pipelines to be re-qualified, this should require samples of as installed coating for qualification testing.

6 PRODUCT COMPOSITION

In the context of CCUS, product composition is assumed to be predominantly carbon dioxide, implicitly > 95% carbon dioxide. With regards to pipeline hydraulic response, a higher fraction of non-condensable components such as N_2, CH_4, H_2 etc. may be allowed without causing operational concerns or affecting fracture arrest criteria. Similar to dense phase, the acceptable levels of impurities for gas phase pipelines need to be established and understood specific to each project. Particular attention should be given to water solubility, effects of impurities, and degradational effects on pipeline material. For projects planning for both gas and dense phase, a dedicated composition for gas- and dense phase respectively may need to be specified to address the difference in water solubility and secondary effects of impurities.

In the context of CCUS, the allowable water content may be limited by other parts of the value chain, e.g. ship transport or end user(s). Water solubility is also affected by the product composition, i.e. the fraction of other impurities. Guidance on water solubility limits for water in 100% gas and dense phase carbon dioxide is given in Figure 5-1 /14/ within typical range in operating temperature and pressures for dense and gas phase respectively. The graph reflects 100% pure CO_2 and should only be taken for information to explain the lower water solubility in gas phase compared to dense/liquid phase. The distinct step in water solubility when going from dense to gas phase should be noted, with a lower solubility in gas phase compared to dense phase. As an example, the water solubility in 100% CO_2 at 15°C drops from approximately 2200 ppm-mol for dense phase to slightly below 700 ppm-mol for gas phase. At a possible minimum (upset) operating temperature for carbon dioxide pipelines of -10°C, the solubility of gas phase CO_2 is slightly below 180 ppm-mol. For reference, the OCAP project operates with a water specification of less than 40 ppm-mol /7/.

For carbon dioxide compositions with other impurities than water, water solubility will differ from 100% CO_2 and should be assessed individually. Water solubility generally decreases with the increasing level of impurities, hence the allowable water content to prevent water drop-out needs to be adjusted accordingly. For a project specific composition, the water solubility should be confirmed either through testing or based on a validated equation of state. Special attention should be given to glycols (TEG/MEG) used for dewatering (drying). In the same way as for dense/liquid phase, consideration should also be given to the solubility of glycols in carbon dioxide for gas phase operation and the effect of glycols on water solubility. Drop-out of a water/glycol phase may form a corrosive condition; this topic is subject to ongoing research.

Similar to dense phase, it is important to pay attention to trace elements of impurities that can form acids that may be highly corrosive even in the gas phase. This relates to chemical reactions between inorganic gas impurities NO_x (NO_2+NO), SO_x (SO_2+ SO_3), H_2S, O_2 and H_2O, forming nitric and sulphuric acid, and solids (e.g. elemental sulphur), with potential for corrosion. Other compounds being a potential source for sulphuric acid formation are COS and CS_2. COS and CS_2 can hydrolyse to H_2S and SO_2, which are reactants for the formation of sulphuric acid. Prediction tools for setting up accurate limits of these gases to avoid precipitation of acids are currently not available. However, there are tools publicly available that can support the decision process for setting up tentative limits of acid producing components that can be used to design an appropriate test program. On the condition that the selected product specification can be sufficiently qualified for the relevant pipeline materials, it is foreseen that internal corrosion can be managed to a level comparable to the current industry experience for CO_2 pipelines. This applies equally to gas phase as for dense phase.

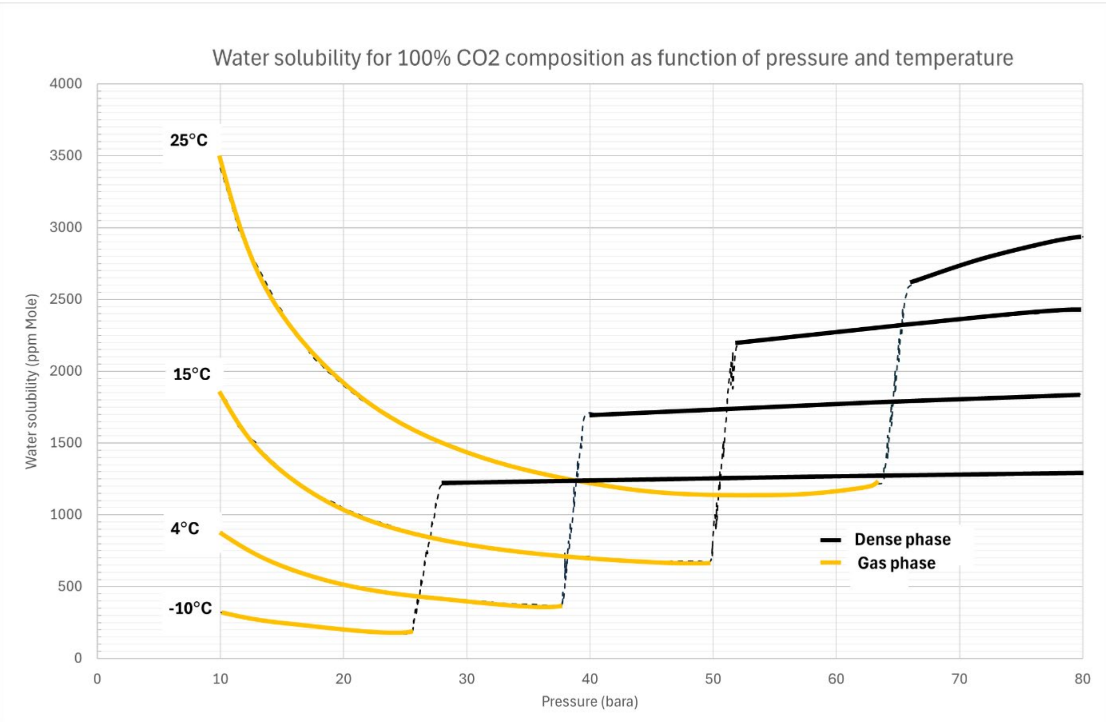

Figure 6-1 Water solubility in 100% CO2 composition for typical gas and dense phase operating conditions; data regenerated from Dynamis project /14/

7 MATERIAL PROPERTIES

Guidance on material selection for carbon dioxide pipelines is provided in DNV-RP-F104 /*1*/ and in a recent study by AMPP /*10*/. For gas phase carbon dioxide products with sufficiently low water content, the reuse of C-Mn steel grade pipelines is considered feasible and typically the preferred choice. Implicitly, material grades traditionally used for e.g. natural gas pipelines would normally be suitable also for transporting CO_2 in gas phase. Special attention should be given to non-metallic materials used in components and compatibility with CO_2. To account for possible upset conditions, non-metallic materials should be qualified both for gas and liquid phase CO_2, including secondary effects of impurities contained in the product stream. If not compatible, the need for seal or component(s) replacement shall be considered.

8 RUNNING FRACTURE

There is a general acceptance within the industry that running fracture needs to be treated differently for carbon dioxide pipelines operated in dense phase compared to natural gas pipelines. Running fracture is associated with running ductile- and brittle- fracture. Methods for how to demonstrate the arrest of running ductile fractures for dense phase pipelines is provided in DNV-RP-F104 /*1*/, however, this is still subject to ongoing research. For carbon dioxide pipelines operated in gas phase, there is a general perception that the physical mechanisms for running fracture are more comparable to natural gas pipelines, where the Battelle Two-Curve Method (BTCM) is considered industry best practice. For gas phase carbon dioxide pipelines, the current working hypothesis is that the Battelle Two-Curve Model (BTCM) is applicable. It should be noted that this is fully validated by testing.

The decompression speed for carbon dioxide is different from natural gas and will be a function of the carbon dioxide composition and the pipeline operating conditions (pressure & temperature) /*11*/.

For comparable gas pipelines, the decompression speed is estimated with appropriate decompression models. A decompression model specific to carbon dioxide is proposed in /*12*/, including model comparison with shock-tube testing. An extract from /*12*/ is given in Figure *8-1*, showing a decompression curve for 100% carbon dioxide, from initial 40 bara and 10°C. The figure includes estimated decompression behaviour based on a simplified isentropic model (green) and an equilibrium model (blue) based on Span-Wagner EoS, compared with measured decompression speed (red). The equilibrium model includes the effect of liquid drop-out when the decompression (due to cooling from gas decompression) reaches the vapour-liquid equilibrium (VLE) curve. This is reflected by the intermediate plateau in the curve. The equilibrium model provides the best approximation of measured data and should be considered the preferred approach. Special attention may be required to repurposing vintage pipelines with potentially low toughness and the effect of a “plateau” in the decompression curve associated with the condensation line for the product composition. The most onerous decompression speed for the specific project should be established based on product composition, operating pressure and temperature, and compared with the fracture propagation speed. With reference to the BTCM, a safety factor should be considered.

For gas phase carbon dioxide pipelines, fracture arrest may be confirmed by one of the following criteria:

- Maximum operating pressure is less than the most onerous (lowest) arrest pressure documented for the line pipe
- Based on BTCM, the predicted decompression speed is higher than the fracture propagation speed for all pressures.

The fracture arrest pressure and propagation speed should be based on documented line pipe material properties (certificates), accounting for material data uncertainty.

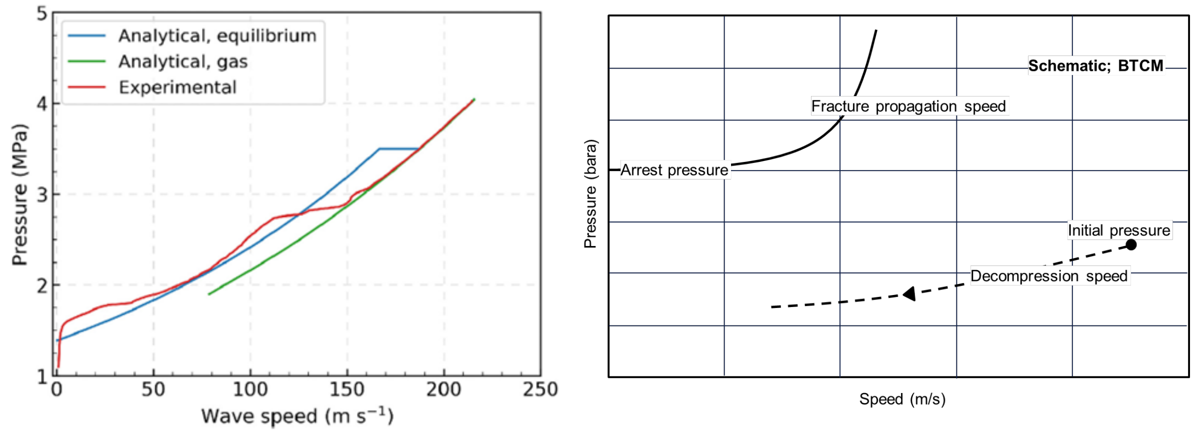

Figure 8-1 Estimated decompression speed (wave speed, m/s) for a 100% carbon dioxide composition, given initial pressure of 40 bara and initial temperature of 10°C /18/ (left). Illustration of BTCM approach (right)

For offshore pipelines designed to DNV-ST-F101 /*14*/, a running fracture for natural gas (methane) pipelines does not need to be considered if the maximum line pipe hoop stress during operation is less than 0.4*SMYS. This criterion may be applicable to re-purposing a submarine pipeline rated for a higher pressure than required for transporting carbon dioxide in gas phase. The above recommendations are confirmed to be in line with recent initiatives, e.g., Pipeline Research Council International (PRCI) /*11*/. For gas phase pipelines planned for operation close to the critical point, special attention should be given to the high gradients and uncertainty in modelling the thermo-physical behaviour.

Running brittle fracture is concerned with the line pipe base material. Brittle fracture arrest shall be assessed with reference to the minimum operating temperature and is normally based on the drop-weight tear test (DWTT) transition temperature being below the minimum design temperature of the pipeline. For gas phase carbon dioxide pipelines similar methodologies as for natural gas pipelines apply. Reference is also given to AS 2885 /*17*/, stating that brittle fracture is not considered likely if the stress is below 85 MPa. Hence, hoop stress below this level may be taken as additional support that the risk of brittle fracture is low.

9 INTEGRITY MANAGEMENT

Significant research is performed to assess corrosion mechanisms for carbon steel materials in a carbon dioxide environment. This includes not only pure carbon dioxide with the presence of a free water phase but also the effects of impurities foreseen to be present in the context of CCUS. For gas phase transport, particular attention should be given to the lower water solubility relative to the product water specification. Similar to dense/liquid phase carbon dioxide pipelines, the primary strategy for managing internal corrosion should be based on strict water control. The industry is currently discussing whether a minimum corrosion allowance may be considered to compensate for any upset conditions and accumulated internal corrosion for pipeline service life. These considerations are equally relevant for gas phase and for dense phase carbon dioxide pipelines.

Sulphide stress cracking (SSC) is the result of metal corrosion and tensile stress in the presence of water and H_2S. If initiated the cracking may be rapid. Similar to internal corrosion, the primary mitigation strategy should be based on water control and to ensure that the material exhibits satisfactory resistance to cracking under the actual environment. Special attention should be given to the carbon dioxide composition, considering the combined effects of different impurities. The effects of different impurities on SSC potential are described in a recent publication from

AMPP /10/ where SSC is addressed in general terms and to a limited extent specifically for gas phase carbon dioxide. The topic is also addressed in the ongoing Safe & Sour JIP /*16*/.

10 SAFETY

Pipeline safety risk is associated with the probability (frequency) of pipeline failure and the associated consequence of loss of containment. Measures to reduce safety risk to an acceptable level cover both pipeline design and operation.

When repurposing existing pipelines for CO_2, the starting point for the safety and integrity assessment, see Figure 2-1 (Step 7), needs to consider the as-designed/installed pipeline and condition prior to a change of product. The potential cost saving of reuse versus newbuild depends strongly on the required pipeline modification work to conclude fit for service.

With regards to safety aspects, gas phase carbon dioxide is foreseen to benefit from being comparable to natural gas pipelines. Among other factors, this relates to the inventory within a given pipeline section being comparable to natural gas for gas phase CO_2 rather than being comparable to dense phase CO_2. For a catastrophic pipeline failure, this has implications for release rates, duration of release and extent of CO_2 concentrations hazardous to people.

Implicitly, code requirements to e.g., pipeline location class, sectioning (distance between line break valves) and requirements to fracture arrest can normally be met without major pipeline modifications (e.g., inclusion of additional line break valves or retrofitting of fracture arrestors). Safety risk for operation with CO_2 in gas phase should still be reflected by appropriate assessments, including consequence and dispersion modelling, see Figure 2-1 (Step 7).

11 SUMMARY AND CONCLUSIONS

Gas phase operation is identified as a possible attractive alternative for the re-use of existing oil and gas pipelines (onshore and offshore) with pressure rating or other features insufficient for dense phase, carbon dioxide gathering networks (particularly in densely populated areas), or for injection in gas phase to depleted formations for permanent storage. Previous screening studies conclude that the existing liquid and gas pipeline systems at large will be technically suitable for transporting carbon dioxide in gas phase. For projects planning for the injection of carbon dioxide into depleted oil and gas formations, the low initial reservoir pressure may dictate an initial phase where the carbon dioxide is transported and delivered to the injection site in gas phase. The obvious downside of gas phase compared to dense phase operation is the lower transport capacity.

The design and operation of carbon dioxide pipelines in gas phase generally pose fewer challenges with regards to technical, regulatory, permitting, and safety aspects compared to dense phase operation. There is a general agreement that pipelines confirmed acceptable for transporting carbon dioxide in dense phase would technically be acceptable also for gas phase operation. For gas phase compared to dense phase, technical aspects relate mainly to the lower potential for running fracture and fewer challenges associated with pipeline operation, including special operations such as pipeline depressurization (for maintenance activities). Special attention to potential negative effects of carbon dioxide on non-metallic materials associated with pipeline components, internal coating, or component seals is still required, similar to dense phase. Due to the significantly lower water solubility in gas compared to dense phase, the maximum allowable water content to prevent water drop-out is stricter for gas phase compared to dense phase.

With regards to regulatory, permitting and safety aspects, gas phase carbon dioxide pipelines are foreseen to benefit from being more comparable to onshore natural gas pipelines with regards to operating pressure and perceived safety risk. Still, carbon dioxide pipelines operated in gas phase share similar challenges for accidental release with regards to near-ground gas dispersion and the extent of hazardous zone as for dense phase carbon dioxide pipelines.

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